

Law as a Recursive Super-Attractor

Subjective Selection, Extended Phenotypes, and Strategic Stability in Legal Evolution

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12 September 2026

Abstract

Human societies repeatedly develop institutions for addressing transgression, compensation, proportionality, authority, and conflict closure. Manvir Singh describes justice institutions as cultural super-attractors and explains recurrent cultural complexity through subjective selection: people generate and retain variants they perceive as useful for satisfying goals. This paper develops a bounded legal-evolutionary extension. It argues that legal ordering is best treated not as one universal super-attractor but as a candidate family of recurrent institutional basins. Subjective selection explains the production and adoption of apparently useful normative variants. Extended Phenotype Theory explains, conditionally, how successful variants construct roles, archives, procedures, and enforcement systems that alter their own transmission environment. Evolutionary Game Theory specifies when associated strategies may stabilize, coexist, or be invaded. Gouldian analysis constrains the synthesis by distinguishing current utility from historical origin and introducing exaptation, structural by-products, contingency, and non-optimality. The resulting process is recursive: institutions not only regulate conduct but also select who may formulate, interpret, and enforce future rules. This second-order selection distinguishes legal institutions from many cultural technologies. The paper separates psychological attraction, cultural adoption, strategic stability, institutional persistence, and legal legitimacy; none entails the next. It proposes seven discriminating predictions and identifies diffusion, power, ecology, objective group benefit, and historical contingency as rival explanations. The model remains a conceptual architecture, not empirical validation of EPT, EGT, Gouldian transfer, or legal universality.

Keywords: subjective selection; cultural super-attractors; legal evolution; extended phenotype; evolutionary game theory; exaptation; historical contingency; institutional feedback; justice; power

1. The recurrent invention of legal order

Human societies differ radically in doctrine, authority, procedure, and sanction. Yet they repeatedly develop structured responses to injury, defection, possession, obligation, and conflict. This recurrence poses a question deeper than the history of any code: why do institutionally organized forms of normative settlement reappear across environments?

Singh's theory of the cultural manifold offers one answer. Complex traditions converge because humans pursue recurrent goals and selectively retain variants that appear to satisfy them. Justice institutions are especially instructive. They commonly impose costs, transfer benefits, calibrate responses to transgressions, and institutionalize procedure. On a relation-restoration account, many such arrangements persist because they appear to repair cooperation while limiting cycles of retaliation.

The legal application is promising but incomplete. A legal system does not merely offer a solution that individuals evaluate. Once established, it changes the available strategies, the distribution of authority, the meaning of evidence, the cost of dissent, and the channels through which future norms become thinkable. Legal institutions therefore participate in the selection process they initially emerged from.

This paper calls that process recursive institutional attraction. The claim is not that law inevitably converges on one form. It is that recurrent social problems create basins of attraction around functional packages, while institutional feedback reshapes the landscape itself.

2. Five phenomena, not one

Theoretical inflation begins when different kinds of success are called “fitness.” This paper separates five phenomena.

Psychological attraction concerns whether a variant seems intelligible, fair, effective, emotionally satisfying, or authoritative.

Cultural adoption concerns whether people copy, teach, invoke, or comply with it.

Strategic stability concerns whether associated conduct resists invasion under specified interactions and payoffs.

Institutional persistence concerns whether roles, records, budgets, procedures, and enforcement reproduce the arrangement through time.

Legal legitimacy concerns whether the institution has authority or can be justified under applicable legal and normative standards.

A harsh punishment may be psychologically attractive yet strategically counterproductive. A doctrine may persist institutionally despite low public appeal. A stable equilibrium may be unjust. A legitimate reform may remain fragile. Treating these outcomes as one evolutionary success variable would conceal rather than explain legal development.

3. Subjective selection of normative technologies

Subjective selection begins with goals and evaluation. Victims may seek repair, vindication, deterrence, recognition, or revenge. Transgressors may seek reintegration, reduced liability, delay, or concealment. Third parties may value peace, cooperation, hierarchy, or public reassurance. These goals overlap imperfectly.

Normative procedures function as cultural technologies when people evaluate them as means to those ends. Compensation, mediation, oath, trial, exile, imprisonment, and public apology offer different packages of perceived efficacy. Feedback is often noisy. A sentence may feel effective even when it has little deterrent effect. A ritual may restore cooperation without producing measurable material change. A highly visible procedure may reassure outsiders while burdening the parties.

Selection is therefore subjective in two senses: goals are psychologically represented, and efficacy is evaluated through limited causal models. But legal subjectivity is never evenly distributed. Actors controlling legislation, adjudication, enforcement, education, media, or archives can introduce and preserve variants that others would reject. A realistic model must weight evaluations by influence.

Let a normative variant v receive evaluations $U_i(v)$ from actors i . Its probability of institutional adoption cannot be modeled as the population mean alone. A minimal expression is:

$$A(v) = F(\sum_i w_i U_i(v), C(v), H, E)$$

where w_i represents influence, $C(v)$ implementation cost, H historical constraints, and E ecological conditions. This is a conceptual relation, not an estimated equation. It clarifies that shared goals do not erase hierarchy.

4. From attractor to super-attractor

A simple attractor is a recurrent variant favored by cognition or transmission. A super-attractor is a complex package whose components interact and recur across varied environments. Legal ordering may qualify only at the level of a functional family.

The candidate package includes:

1. classification of a transgression or dispute;
2. identification of parties and standing;
3. an authorized process of narration or proof;
4. a response involving cost, transfer, restoration, or status change;
5. proportionality or another calibration rule;
6. closure, memory, or a rule for future cases.

No claim is made that every society contains every element, or that each element has a single function. The empirical question is whether these components co-occur more often than diffusion, shared ancestry, ecology, or coding choices would predict.

The expected convergence is functional rather than morphological. Blood compensation, mediation, restorative conference, civil damages, and criminal sentence look different. They may nonetheless occupy neighboring regions in a design space because they combine recognition, calibrated response, and conflict management.

4.1 The unit of analysis

The relevant unit is neither an isolated rule nor an entire legal system. A rule is often too small because its effects depend on procedure, personnel, records, sanctions, and shared classifications. A legal system is often too large because it contains mutually inconsistent arrangements selected under different conditions. The proposed unit is the **normative-institutional package**: a partially integrated configuration of classifications, procedures, roles, material supports, and expected responses organized around a recurrent coordination problem.

This intermediate unit permits comparison without assuming that “law” is culturally uniform. A compensation schedule, for example, may be coupled to kin responsibility, ritual recognition, a neutral intermediary, and a closure rule. A modern damages regime may combine individualized standing, documentary evidence, professional representation, monetary valuation, and appellate review. They are not instances of the same doctrine. They may nevertheless be compared as packages that transform an alleged injury into an authorized sequence of classification, proof, response, and closure.

The package is also the candidate replicating structure for the EPT extension. What must recur is not every word or institutional detail, but enough organized information to reconstruct the relevant roles and decision sequence. This reconstruction can occur through teaching, imitation, professional training, templates, precedents, or material infrastructure. Whether that continuity amounts to cultural inheritance is an empirical question requiring a traceable genealogy. Similarity alone is insufficient.

4.2 Levels of attraction

Attraction may operate at several levels. Individuals may prefer a procedure because it promises vindication or predictability. Coalitions may support it because it protects status or lowers internal conflict. Organizations may reproduce it because budgets, careers, and records are built around it. Politics may retain it because it contributes to governability or interstate credibility. These pressures need not point in the same direction.

A procedure can therefore be attractive to the institution while unattractive to most people subject to it. Conversely, a publicly compelling reform can fail because it supplies no durable role structure, funding mechanism, archive, or enforcement pathway. The super-attractor claim becomes meaningful only if it specifies the level at which attraction occurs and the transmission channel through which it affects persistence.

5. Extended phenotype as recursive infrastructure

Subjective selection explains why people adopt a procedure. It does not fully explain why the procedure later shapes its adopters.

EPT adds a conditional mechanism. A normative pattern can produce external structures that improve its reproduction: professional roles, records, buildings, curricula, forms, precedents, budgets, and technical standards. These structures make some interpretations cheap and authoritative while making alternatives costly or invisible.

The recursion is time-indexed:

$$V_t \rightarrow I_t \rightarrow \{P_{t+1}, K_{t+1}, G_{t+1}\} \rightarrow V_{t+1}$$

V is the distribution of normative variants. I is institutional infrastructure. P is the payoff environment, K the available knowledge and categories, and G the gatekeeping structure. Institutions at time t alter the conditions under which variants are generated and selected at t+1.

This model avoids attributing intentions to norms. Human actors build and maintain institutions. EPT earns its place only if the external effect systematically changes differential transmission of identifiable variants.

5.1 What EPT adds, and what it does not

The extended-phenotype analogy is useful only under disciplined conditions. First, there must be an identifiable informational variant or package. Second, the package must contribute causally to an external structure. Third, that structure must affect the probability that the package, or a genealogically connected descendant, is retained or reconstructed. Fourth, the account must outperform a simpler description in terms of organizational inertia, sunk cost, or coercion.

Consider precedent. A judicial formulation may be copied because later actors regard it as persuasive. But once embedded in citation practices, databases, procedural burdens, and professional education, it also changes the search environment in which later lawyers formulate claims. The external architecture can amplify some formulations and suppress others. This is a candidate extended-phenotypic loop because the doctrinal variant helps construct a transmission environment favorable to its descendants. It is not enough that the precedent has consequences or that institutions exist around it.

The same test applies to codes, forms, jurisdictional rules, registries, and professional categories. They become relevant to EPT when they carry forward organized information and modify differential

reproduction. If a courthouse merely symbolizes authority without affecting transmission, EPT may add little. If a filing system determines what can be seen, compared, appealed, and cited, the mechanism is stronger.

5.2 Four feedback pathways

The recursive infrastructure can operate through at least four pathways.

Cost feedback: standardized forms, trained personnel, and established venues reduce the marginal cost of using the incumbent package while raising the cost of alternatives.

Epistemic feedback: archives and authorized classifications determine which events become legible as injuries, evidence, precedents, or exceptions.

Strategic feedback: enforcement capacity and expectations alter the payoffs of compliance, evasion, retaliation, and institutional participation.

Constitutive feedback: legal categories reshape roles and preferences. An actor may learn to understand a grievance as a contractual breach, discriminatory act, property invasion, or public-law claim because the institution supplies those categories.

These pathways are analytically distinct. Cost feedback can occur without preference change. Epistemic feedback can change available reasons without changing sanctions. Strategic feedback can stabilize conduct without legitimacy. Constitutive feedback is the strongest and most difficult claim because it requires evidence that institutional categories helped form, rather than merely record, actors' goals.

6. Gouldian constraints: contingency, exaptation, and imperfect form

The super-attractor hypothesis can easily become adaptationist. If recurrent institutions are identified by the problems they solve, researchers may be tempted to infer that every component was selected for its present function or that convergence implies optimization. Stephen Jay Gould's critique of adaptationist explanation provides a necessary correction.

First, present utility does not establish historical origin. A legal structure may be useful today even though it arose for a different purpose. Gould and Vrba's concept of **exaptation** is therefore central to legal evolution. Archives created for taxation or dynastic administration can later support accountability. Procedural forms developed to manage official workflow can become protections against arbitrary decision. Corporate forms, jurisdictional devices, and inherited remedies can be recruited for problems their designers never anticipated. The current contribution of a feature must be separated from the process that produced it.

Second, not every institutional component is a direct solution. Some features may be structural by-products, comparable in analytical role to the spandrels discussed by Gould and Lewontin. A classification, exception, or procedural residue can emerge because of the way a larger system is organized and only later acquire an independent use. Calling every surviving feature adaptive would hide these dependencies. In legal analysis, the spandrel concept should function as a hypothesis about origin, not as a label for whatever appears irrational.

Third, recurrent attraction does not eliminate contingency. If comparable social problems were replayed, some functional families of response might reappear, but their doctrinal and institutional forms could differ radically. Existing institutions evolve from available materials, inherited categories, prior

settlements, and accidental sequences. The super-attractor supplies a constrained region of possibility, not a predetermined destination.

Fourth, institutional change may be temporally uneven. Long periods of relative stability can be interrupted by concentrated reorganization during crises, transitions, or technological shocks. This pattern may invite comparison with punctuated equilibrium, but the analogy must remain bounded. A legal episode of rapid change is not evidence that the biological model of Eldredge and Gould has been transferred. The relevant empirical question is whether institutional feedback and constraints generate measurable stasis and clustered change more successfully than gradualist alternatives.

Finally, causal influence can be hierarchical without proving selection at every level. Organizations, constitutions, and legal traditions can constrain lower-level variation, alter exposure, and filter survival. That does not by itself make them Darwinian individuals or units of selection. A multi-level account must distinguish organizational hierarchy, downward constraint, and genuine selection among higher-level entities.

The resulting synthesis is deliberately non-optimal. Subjective selection helps explain why actors retain solutions they perceive as useful. Gouldian analysis explains why those solutions are built from inherited materials, why by-products survive, why present functions may differ from origins, and why history does not reliably converge on the best available design. A recursive super-attractor is therefore not an institutional optimum. It is a recurrent basin whose concrete contents remain contingent, constrained, and open to exaptation.

7. Evolutionary games and institutional basins

EGT contributes a different layer. It asks when strategies associated with a normative arrangement remain stable against alternatives.

Suppose a population confronts recurrent harms. Victims choose retaliation or institutional repair. Transgressors choose compensation or resistance. Third parties choose mediation, enforcement, or inaction. The value of future interaction, enforcement credibility, information quality, and repair cost influence the equilibrium.

Under high interdependence and credible mediation, institutional repair may outperform escalating retaliation. Under low enforcement or severe power asymmetry, denial and coercion may dominate. Under noisy information, costly signaling through severe punishment may persist even without strong deterrence.

These are model-dependent propositions. Calling a long-lived doctrine an ESS without specifying population, strategies, interactions, payoffs, and updating would be a category error. The safer term “institutional basin” describes path-dependent attraction without importing formal stability prematurely.

7.1 A minimal coordination model

Let a population repeatedly face a harmful event. An affected party chooses between direct retaliation R and institutional recourse I . A responding party chooses resistance D or repair C . Relevant third parties choose enforcement E or inaction N . Expected payoffs depend on the probability q that an authorized process correctly classifies the event, the probability e that a decision is enforced, the future-interaction parameter δ , the cost c of procedure, and the distribution of power h .

The institutional pathway becomes comparatively attractive when the expected value of recognized repair, future cooperation, and reduced retaliatory loss exceeds procedural cost and the expected gains from direct action. In deliberately schematic form:

$$EU(I) - EU(R) = q[eB + \delta F] - c - H(h)$$

B represents the value of the authorized response, F the expected value of preserved future interaction, and H(h) the disadvantage produced by asymmetric power. The expression is not a universal payoff function. It identifies observable dimensions that a later model would need to operationalize. In some environments, greater state capacity raises e and favors institutional recourse. In others, concentrated power raises H(h) because enforcement selectively protects dominant actors.

This ambiguity is important. The same institutional growth that improves credible enforcement can deepen exclusion. There is no monotonic relationship between institutionalization and justice.

7.2 From local equilibrium to basin depth

An equilibrium describes mutual best responses under specified conditions. A basin describes the set of starting states and disturbances from which a system tends to return to a configuration. For legal evolution, basin depth may depend on more than payoffs at one moment. Archives, professional investment, jurisdictional boundaries, and public expectations can make departure costly and return likely.

A conceptual persistence function is:

$$P(v, t+1) = G[A(v,t), S(v,t), X(v,t), L(v,t), Z(t)]$$

A denotes adoption pressure, S strategic reinforcement, X transmission infrastructure, L coercive or political support, and Z exogenous conditions. No additive or linear form is presumed. The purpose of the expression is to prevent a single observed outcome, persistence, from being attributed automatically to subjective selection or strategic stability.

The distinction also clarifies path dependence. Two societies with similar current payoffs may occupy different basins because their archives, professional structures, and authorized narratives differ. Conversely, a severe external shock can move a system across a boundary even when the incumbent institution remains locally stable.

7.3 Multi-level conflict

Strategies may be selected at individual, organizational, coalition, and polity levels. A procedure that is costly for individual claimants may benefit a professional organization. A rule that stabilizes a ruling coalition may reduce aggregate cooperation. A reform that improves group performance may threaten incumbent gatekeepers and fail to reproduce.

The framework therefore rejects any inference from institutional persistence to social benefit. Multi-level selection is a possible explanation only when units, variation, differential persistence, and transmission are specified. Where those conditions cannot be shown, the safer account is conflict among actors operating through institutional constraints.

8. Second-order selection

Law becomes distinctive when it governs selection itself. A legal system determines:

- who may initiate a claim;

- which facts enter the record;
- who counts as an expert;
- which interpretations bind later decision makers;
- which sanctions are available;
- which forms of change are legally cognizable.

The first-order institution selects conduct. The second-order institution selects the selectors, evidence, and mutation channels of future law. Constitutional amendment rules, jurisdiction, standing, precedent, professional accreditation, and administrative procedure are examples.

This feature supports the description of law as recursive. It also explains why subjective usefulness can become endogenous. People learn to perceive disputes through categories the institution supplies. Property, injury, personhood, capacity, and public interest are not merely solutions to pre-institutional goals. They reorganize those goals.

8.1 Mutation channels and selector control

Every legal order contains channels through which variants may be proposed and recognized. Legislation, adjudication, contract, administrative rulemaking, custom, and professional interpretation are not simply sources of rules. They are mutation channels with different entry costs, rates, and gatekeepers.

A constitutional amendment rule limits which high-level variants can enter. Standing rules determine which grievances receive an authoritative hearing. Evidentiary rules determine which descriptions of reality survive into the record. Precedent determines how much inherited information constrains a new decision. Professional accreditation determines who may generate legally consequential interpretations.

Second-order selection occurs when the incumbent package influences these channels. A doctrine that defines its own review standard, a profession that controls admission, or an agency that specifies the evidence by which its own performance will be judged participates in selecting the selectors. This does not prove self-serving design. It identifies a feedback structure in which the object of selection partly configures future selection.

8.2 Recursive attraction and lock-in

Recursion can deepen an institutional basin, but recursion and lock-in are not synonyms. Feedback may improve learning, permit correction, and stabilize beneficial expectations. Lock-in requires a stronger showing: switching costs, increasing returns, constrained alternatives, or self-reinforcing authority that preserves a package despite credible superior options.

The empirical signature of recursion is therefore not mere longevity. It is a sequence: adoption of a package, construction or reconfiguration of selection infrastructure, measurable alteration of later variation or evaluation, and differential retention connected to that alteration. Without temporal evidence for those links, the recursive-super-attractor description remains a hypothesis.

9. Rival explanations

The super-attractor hypothesis must compete with at least five rivals.

Common descent and diffusion: recurrent legal forms may spread historically rather than re-emerge independently.

Ecological constraint: similar subsistence, demography, or conflict patterns may generate similar institutions without shared psychology doing the main work.

Objective group benefit: institutions may persist because they improve cooperation or group competition, regardless of individual perceptions.

Power and coercion: elites may impose arrangements that serve their interests and shape subordinate preferences.

General cognitive attraction: memory, learning, causal intuitions, and emotional salience may stabilize forms without goal-directed evaluation.

The preferred explanation will often be plural. The scientific task is to estimate which mechanism explains which feature under which conditions.

9.1 The circularity problem

An attractor account risks becoming circular: an institution is said to persist because it is attractive, and its persistence is offered as evidence of attraction. Subjective selection does not escape this problem merely by invoking perceived usefulness. Independent measures are required. Researchers must estimate actors' goals and evaluations separately from adoption outcomes, then test whether those evaluations predict variation or retention after accounting for power, exposure, cost, and inheritance.

The same caution applies to integration. If every co-occurring institutional feature is defined as part of the package after observation, the super-attractor claim cannot fail. Components and expected dependencies must be specified before testing. Holdout societies, periods, or institutional domains are necessary to distinguish explanation from retrospective redescription.

9.2 Functional equivalence and observer bias

Cross-cultural comparison can manufacture convergence by coding unlike practices at an excessively abstract level. Nearly any response to conflict can be redescribed as “closure,” “repair,” or “calibration.” The solution is not to abandon functional comparison but to use multiple resolutions. Researchers should record local categories and meanings, observable sequences, material transfers, participation rights, enforcement mechanisms, and proposed higher-level functions separately.

Convergence is more credible when independent coders using preregistered criteria identify the package, when local accounts support the proposed function, and when the relationship survives controls for descent and diffusion. Disagreement among coding levels is itself evidence about the limits of the super-attractor category.

9.3 Power as a constitutive rival

Power is not merely a variable that distorts otherwise neutral selection. It can determine the goals that institutions are designed to satisfy, whose evaluations count, which alternatives are visible, and which costs remain externalized. An institution imposed through conquest can later become psychologically familiar and strategically stable. Present attraction would not establish an origin in shared problem-solving.

The framework therefore requires separate explanations of emergence and persistence. Subjective selection may explain an innovation's origin, coercion its diffusion, infrastructure its persistence, and later socialization its apparent naturalness. Conversely, a coercively introduced institution may be repurposed through local evaluation. A plural sequence is more plausible than a single master cause.

10. Discriminating predictions

7. Perceived efficacy should predict adoption more strongly when objective feedback is delayed or ambiguous.
8. Institutions should track the evaluations of high-influence actors more closely where enforcement and communication are concentrated.
9. Functionally similar justice packages should recur more than identical doctrinal forms after controlling for historical relatedness.
10. Institutional variants that create archives and authorized interpreters should show greater persistence, conditional on prior power and age.
11. Changes in institutional categories should precede measurable changes in how actors formulate goals and grievances.
12. Restoration-oriented procedures should be more stable where future interdependence is high; deterrence-oriented procedures should increase with anonymity and lower repeat interaction, subject to state capacity.
13. Removing a component of an integrated legal package should degrade function more than removing an incidental symbol. This tests functional integration rather than mere co-occurrence.

These predictions are discriminating only if their rivals are specified. Prediction 1 competes with objective-performance and prestige-bias accounts. Prediction 2 competes with population-average preference models. Prediction 3 competes with diffusion and common descent. Prediction 4 competes with age, wealth, and coercive-capacity explanations. Prediction 5 targets constitutive feedback rather than simple preference aggregation. Prediction 6 supplies conditions under which different strategic packages should vary. Prediction 7 tests whether the proposed super-attractor is an integrated design rather than a list assembled by the observer.

Evidence against these predictions should narrow the theory. If perceived efficacy adds nothing once objective performance and prestige are controlled, the subjective-selection component weakens. If archives and authorized interpreters do not predict persistence beyond institutional age and resources, the EPT mechanism loses support. If interaction structure fails to predict strategy distributions, the proposed EGT bridge should be rejected or redesigned. The architecture is useful only if its layers can fail independently.

11. Legal and normative boundary

Evolutionary recurrence has no automatic legal authority. A practice can be widespread, stable, and subjectively appealing while violating equality, dignity, or due process. Conversely, a justified institution may be historically unusual and strategically fragile.

The framework therefore stops before normative judgment. Positive-law validity requires jurisdiction-specific sources. Legitimacy requires defensible normative premises. Liability requires rules connecting conduct, causation, fault, authority, and remedy. Evolutionary explanation supplies none of these by itself.

12. Research design

The proposed comparative study would construct a stratified sample of dispute institutions across societies and historical periods. It would code functional components, actor goals, perceived and objective efficacy, power concentration, transmission infrastructure, interaction structure, and persistence.

Linguistic and geographic relatedness would address diffusion. Ecological and political variables would address alternative pressures.

Process tracing would test the EPT loop in selected cases: did a normative variant create infrastructure, did that infrastructure alter available interpretations or payoffs, and did those changes affect later transmission? Formal models would be reserved for cases with specifiable strategies and interactions.

No single dataset can establish the entire architecture. The design is modular. A failed prediction should eliminate or narrow the corresponding mechanism without requiring abandonment of every layer.

12.1 Comparative module

The first module would sample societies and historical periods using criteria fixed before coding. It would avoid treating modern states as the default form and would preserve distinctions among customary, religious, kin-based, administrative, civil, and criminal arrangements. The dependent variables would include component co-occurrence, adoption, persistence, and reconstruction after disruption. Geographic proximity, linguistic ancestry, known contact, subsistence, population density, warfare, inequality, and state capacity would serve as controls or stratification variables.

The central test is not whether all societies possess “law.” It is whether specified functional packages recur with greater regularity and integration than rival models predict. Negative cases are essential: recurrent conflicts managed without an authorized third party, packages lacking closure, institutions that disappear despite perceived usefulness, and durable arrangements that most participants judge ineffective.

12.2 Process-tracing module

The second module would select cases with sufficient temporal documentation to reconstruct the feedback sequence. For each case, researchers would identify the initial normative variant, the actors and goals involved in adoption, the infrastructure subsequently created, and later changes in payoffs, categories, gatekeeping, or transmission. Competing process predictions would be stated in advance.

Evidence for recursive EPT would require more than institutional growth after adoption. It would require a plausible causal link from the variant to the infrastructure and from that infrastructure to differential reproduction. Evidence of institutional persistence caused only by repression or fiscal dependence would support a rival mechanism. Mixed cases should be coded as mixed rather than forced into a single explanation.

12.3 Experimental and computational module

Laboratory, survey, and simulation studies could isolate limited mechanisms. Participants might choose among dispute procedures while perceived efficacy, objective efficacy, visibility, power asymmetry, and future interaction are varied independently. The design should distinguish stated approval from behavioral adoption and should measure whether institutional categories change subsequent descriptions of the dispute.

Agent-based or evolutionary simulations could explore whether packages combining classification, enforcement, memory, and closure occupy wider basins than isolated rules. Such simulations would establish consequences of assumptions, not evidence that historical societies followed the model. Parameters should be linked to observed ranges where possible, and sensitivity analysis should identify whether claimed patterns depend on narrow choices.

12.4 Falsification and stopping rules

The recursive-super-attractor proposal should be abandoned in its strong form if preregistered coding fails to find recurrent integrated packages after correction for relatedness and diffusion. The EPT extension should be abandoned if transmission infrastructure does not mediate or moderate differential persistence in appropriately traced cases. The EGT layer should be restricted to local models if no stable relationship appears between interaction structure and strategy distribution.

More modest findings would justify narrower conclusions. Recurrent psychological attraction without institutional feedback would support subjective selection but not recursion. Infrastructure-mediated persistence without independent attraction would support an institutional inheritance account but not a super-attractor claim. Strategic stability without legitimacy would remain a positive explanation only. These stopping rules prevent the framework from surviving every possible result by changing vocabulary.

13. Scope, contribution, and epistemic status

The contribution of this paper is architectural. It connects three explanatory traditions while preserving their distinct dependent variables. Subjective selection concerns evaluated usefulness in cultural production and retention. EPT concerns externally organized transmission feedback. EGT concerns frequency-dependent strategic dynamics under specified conditions. Legal theory adds the separate problems of validity, authority, interpretation, and justification.

The proposed synthesis does not claim that genes encode legal rules, that institutions literally reproduce as organisms, or that every persistent rule is an extended phenotype. It does not claim that pre-linguistic coordination was already law in the full doctrinal sense. It does not infer ought from survival, legitimacy from equilibrium, or universality from recurrence.

Its narrower thesis is that some legal packages may be explained better when their psychological attraction, strategic environment, transmission infrastructure, and control over future variation are analyzed together. The phrase **recursive super-attractor** names this candidate configuration. Whether it identifies a genuine cross-cultural regularity depends on evidence not supplied by this conceptual paper.

14. Conclusion

Law may have begun as a response to recurrent problems of coordination, injury, retaliation, and repair. Once institutionalized, however, it ceased to be merely one solution among others. It began to organize the environment in which future solutions could emerge.

This is the central implication of the recursive-super-attractor hypothesis. Legal institutions do not only stabilize expectations or alter strategic payoffs. They determine who may speak with authority, which injuries become cognizable, what counts as evidence, which interpretations enter institutional memory, and which normative variants remain available for future selection. Law is therefore not simply selected within a cultural environment. It progressively participates in constructing that environment.

Subjective selection helps explain why certain normative technologies are produced and retained. EPT identifies the conditions under which those technologies build infrastructures that favor their own transmission. EGT clarifies when the strategies they organize become stable, coexist with rivals, or remain vulnerable to invasion. Gouldian analysis supplies the indispensable negative lesson: recurrence is not optimality, present utility is not original function, persistence is not adaptation, and structured change remains historically contingent. None of these mechanisms establishes legal validity or justice. Together,

however, they disclose a possibility obscured by accounts that treat law only as command, language, or deliberate design.

The deepest continuity between pre-linguistic coordination and legal order may thus lie neither in rules nor in sanctions, but in the progressive externalization of social expectations into environments that constrain subsequent choice. Language made those expectations more abstract, portable, and contestable. Institutions gave them memory, authority, and force. Exaptation repeatedly recruited inherited structures for new tasks. Recursion began when those institutions acquired the capacity to govern their own conditions of variation.

The claim remains falsifiable and may ultimately require substantial restriction. Yet if it survives comparative and historical testing, law should no longer be understood merely as a cultural product shaped by selection. It must also be understood as a selective architecture: an imperfect human construction, assembled from contingent materials, that helps determine which forms of normative life can be imagined, transmitted, contested, and made real.

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